3 FLOW CONTROL

3.1 Delay in Computer Network

(1) Transmission Delay (T_d)
(2) Propagation Delay (P_d)
(3) Queuing Delay (Q_d)
(4) Processing Delay (P_{rd})

3.2 Transmission Delay

Amount of time taken to transfer a packet on to the outgoing link is called as Transmission delay.

$$\text{Transmission delay}\,(T_d) = \frac{\text{Length of packet}}{\text{Bandwidth}}$$

$$T_d = \frac{L}{B}$$

3.3 Propagation Delay

Amount of time taken to reach a packet from one (sender) point to another (receiver) point is called as propagation delay.

$$\text{Propagation delay}\,(P_d) = \frac{\text{distance}}{\text{velocity}}$$

$$P_d = \frac{d}{v}$$

3.4 Stop wait Protocol

3.4.1 Sender Side Rule

Rule 1 : Sender can send one data packet at a time.
Rule 2 : Sender can send the next data packet only after receiving the ACK of the previous packet.

3.4.2 Receiver Side Rule

Rule 1: Receiver will receive and consume the data packet.
Rule 2 : After consuming the data packet, Ack need to be sent.

3.5 Efficiency OR Line utilization OR Link utilization OR Sender utilization

$$\text{Efficiency} = \frac{\text{Useful time}}{\text{Total time}}$$

$$\text{Efficiency} = \frac{T_d(\text{frame})}{T_d(\text{frame}) + 2*P_d + Q_d + P_{rd} + T_d(\text{ACK})}$$

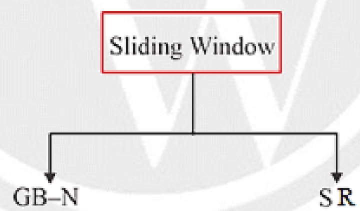
3.6 Throughput Or Effective Bandwidth Or Bandwidth Utilization Or Maximum Data Rate Possible

$$\text{Throughput} = \frac{\text{Length of the Frame}}{\text{Total time}}$$

$$\boxed{\text{Throughput} = \frac{L}{T_d(\text{frame}) + 2*P_d + Q_d + P_{rd} + T_d(\text{ACK})}} \quad \text{OR} \quad \boxed{\text{Throughput} = \eta * B}$$

3.7 Sliding Window

In the sliding window concept instead of sending one packet and wait for the acknowledgement, we send 'w' packet and wait for the Acknowledgement. Where 'w' is the sender window size.



3.7.1 GB–N(N>1)

1. In the GB – N the sender window size is N itself.
2. In the GB-N the receiver window size is equal to one always.

Note:
(1) Out of order packet is not received by Receiver.
(2) Timer is maintained only for the first frame (Rightmost) in window because if its timer expires then sender assume that rest of the frame are also not received by receiver (because out of order delivery is rejected).

Note:
GB–N uses cumulative Acknowledgement and Acknowledgement number defines the number of the next expected frame.

Ack timer < Time out timer

Window Receiver (W_R) size:

In the GB-N the window receiver size is equal to one always irrespective of window sender size (Wr=1).

Window Sender (W_S) Size:

Window sender size is calculated based on the following formula:

$Ws + W_R \leq$ Available Sequence Number

$Ws + 1 \leq$ Available Sequence Number

$Ws \leq$ Available Sequence Number – 1

3.7.2 Efficiency and Throughput in GBN

$$\text{Efficiency} = \frac{\text{Useful time}}{\text{Total time}}$$

$$\boxed{\text{Efficiency} = \frac{N*T_d(\mathit{frame})}{T_d(\mathit{frame})+2*P_d+Q_d+P_{rd}+T_d(ACk)}}$$

$$\text{Throughput} = \frac{N*\text{Length of Frame}}{\text{Total time}}$$

$$\boxed{\text{Throughput} = \frac{N*\text{Length of Frame}}{T_d(\mathit{frame})+2*P_d+Q_d+P_{rd}+T_d(ACk)}} \quad \text{OR} \quad \boxed{\text{Throughput} = \eta * B}$$

3.8 Selective Repeat ARQ

3.8.1 Selective Repeat/Selective Reject ARQ

(1) In SR Protocol window sender size is equal to window receiver size ($W_S = W_R$).

(2) SR Protocol uses independent acknowledgement, and acknowledgement number defines number of error free packet received.

(3) SR receiver can receive out of order packet but packets are delivered to upper layer in sorted order.

(4) In SR protocol searching and sorting logic is required. Searching is done by sender and sorting is done by receiver.

(5) Timer is maintained for each and every frame in the window at sender side.

(6) For 1st out of order delivery or if packet received is corrupted then Negative acknowledgment (NACK) for respective packet is sent by receiver to sender.

(7) When sender receive NACK 3 then it will search in the window for packet 3 & immediately packet 3 is retransmitted even though its timer is not expired.

3.8.2 Efficiency and Throughput of SR

$$\text{Efficiency} = \frac{\text{Useful time}}{\text{Total time}}$$

$$\boxed{\text{Efficiency} = \frac{Ws \times T_d(\textit{frame})}{T_d(\textit{frame}) + 2*P_d + Q_d + P_{rd} + T_d(ACK)}}$$

$$\boxed{\text{Throughput} = \eta * B}$$

$$\text{Throughput} = \frac{W_s * \text{Length of the frame}}{\text{Total time}}$$

$$\boxed{\text{Throughput} = \frac{W_s * \text{Length of the frame}}{T_d(\textit{frame}) + 2*P_d + Q_d + P_{rd} + T_d(ACK)}}$$

3.9 Comparison among Stop and Wait, GBN and SR protocols

	Stop & wait	**GBN**	**SR**
Efficiency	$\eta = \frac{\text{Useful time}}{\text{Total time}}$ or $\eta = \frac{T_d(\textit{frame})}{\text{Total time}}$	$\eta = \frac{\text{Useful time}}{\text{Total time}}$ or $\eta = \frac{N*T_d(\textit{frame})}{\text{Total time}}$	$\eta = \frac{\text{Useful time}}{\text{Total time}}$ or $\eta = \frac{W_s*T_d(\textit{frame})}{\text{Total time}}$
Throughput	$\frac{\text{Length of frame}}{\text{Total time}}$ or $\eta * B$	$\frac{N*\text{Length of the frame}}{\text{Total time}}$ or $\eta * B$	$\frac{W_s*\text{Length of the frame}}{\text{Total time}}$ or $\eta * B$
Buffer	1 + 1	N + 1	N + N
Seq No.	2	N + 1	2N
Seq. No. = K bit		$W_s = 2^K - 1$ $W_R = 1$	$W_S = 2^{K-1}$ $W_R = 2^{K-1}$

$$\boxed{\text{RTT or Total Time} = T_d(\textit{frame}) + 2*P_d + T_d(ACK) + P_{rd} + Q_d}$$

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